20 Most Important Questions

1. **For the silicon diode the reverse saturation current is $20\mu A$ at 27 degree centigrade. What will be the value of reverse saturation current at 87 degree centigrade?**

   Solution:
   Given,
   Reverse saturation current at 27 degree centigrade $(I_S) = 20\mu A$
   Reverse saturation current at 87 degree centigrade = ?

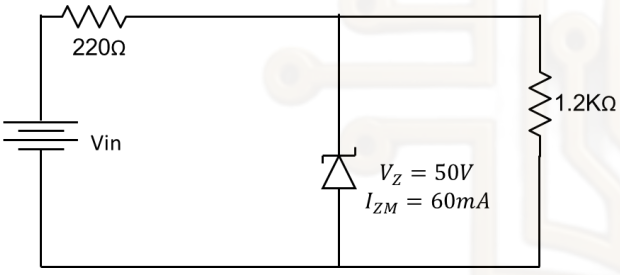
   We know that reverse saturation current doubles at every 10 degree change in temperature.

   Total change in temperature = $87 - 27 = 60$

   Therefore, reverse saturation current doubles 60/10=6 times

   Now, Reverse saturation current at 87 degree centigrade = $2^6 \times 20\mu A = 64 \times 20 = 1280\mu A$

2. **Determine the range of values of Vin that will maintain the Zener diode in the ON state. Also find the maximum power that can be dissipated to the diode.**



   Solution:

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   Given,
   Series resistance (R_s) = 220Ω
   Load resistance (R_L) = 1.2KΩ
   Zener Voltage (V_Z) = 50V
   Maximum Zener current (I_{ZM}) = 60mA

   Range of input voltage (V_{in}) = ?

   Current in load resistance (I_L) = $\frac{50}{1.2K} = 41.66\ mA$

   For maximum input voltage that can switch on the diode,

   Current through series resistance (I_S) = 60 + 41.66 = 101.66mA
   Therefore, voltage in series resistance (V_s) = $101.66 \times 220 = 22.365V$

So, maximum voltage that can be applied (V_{in}) = $50 + 22.365 = 72.365V$

For, minimum input voltage that can switch on the diode,
Current through series resistance (I_S) = 41.66 mA [Almost zero current flows through diode]
Therefore, voltage in series resistance (V_s) = $41.66 \times 220 = 9.1652V$
So, minimum voltage that can be applied (V_{in}) = $50 + 9.1652 = 59.1652V$

3. What are VI characteristic of diode? What happens when a forward biased diode is suddenly reverse biased? Explain with necessary diagrams.

Solution:

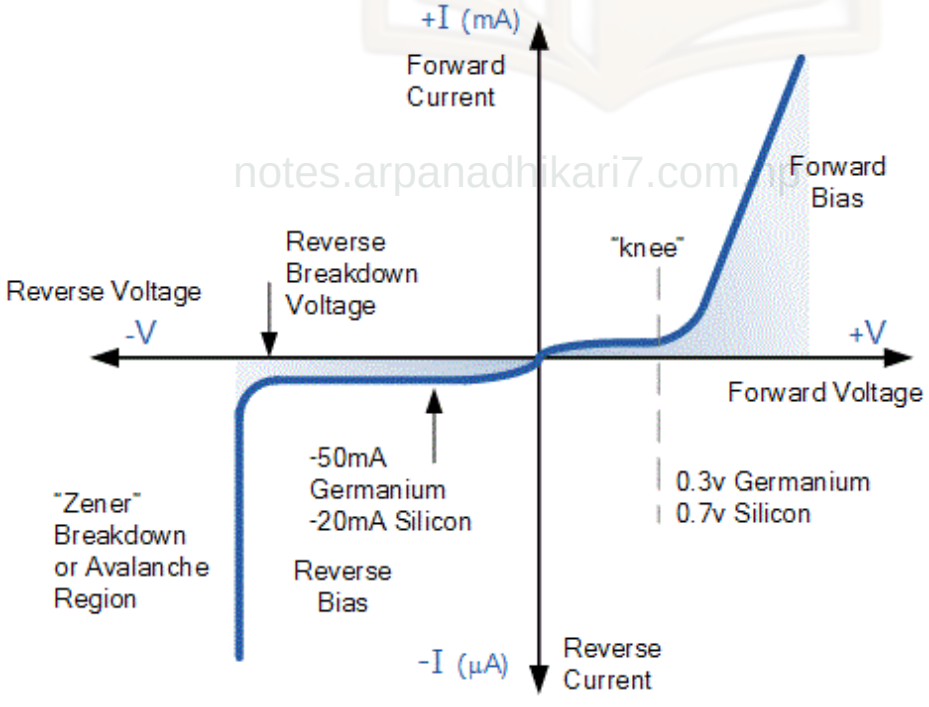
The VI characteristics of a diode refer to the relationship between the voltage applied to the diode and the current that flows through it. It is a graph that tells you how the diode behaves under different voltage conditions.

Here is a breakdown of the key points:

Regions of operation:

Zero bias: When no voltage is applied, the diode does not conduct any current.

Forward bias: When the positive voltage is applied to the p-type side and the negative voltage to the n-type side, the current starts flowing through the diode. Initially, the current increases slowly, then rises sharply as the voltage increases further. This is because the applied voltage overcomes the internal barrier in the diode.



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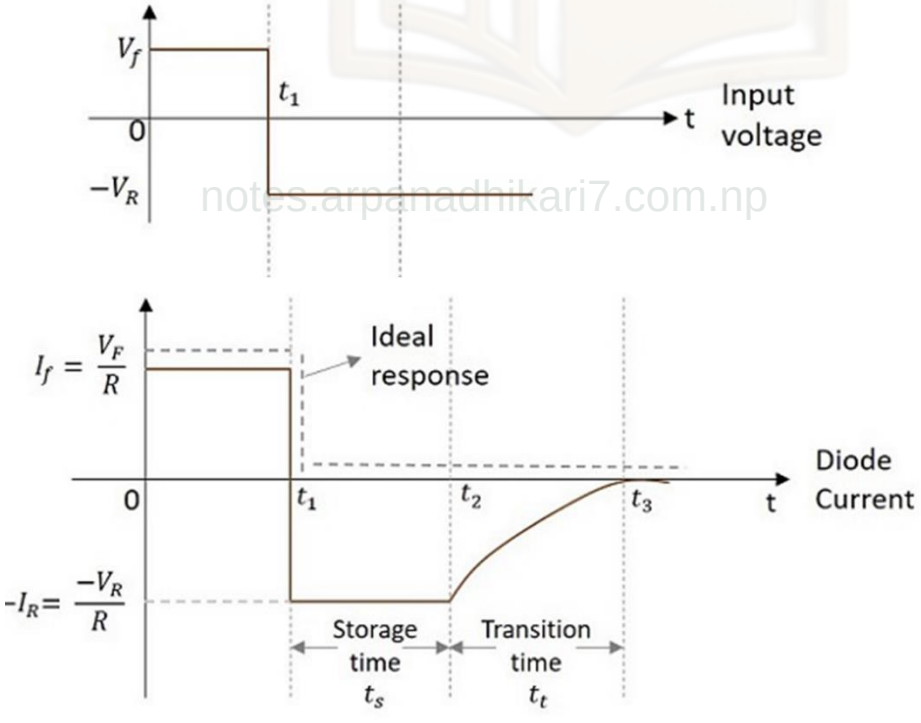
Reverse bias: When the positive voltage is applied to the n-type side and the negative voltage to the p-type side, the diode ideally blocks any current flow. However, a small leakage current might still exist.

Key features of the VI curve:

Knee voltage: The minimum voltage required to forward bias the diode and initiate significant current flow. For silicon diodes, it is around 0.7 V, while for germanium diodes, it is around 0.3 V.
Reverse leakage current: A small current that flows in the reverse bias region, typically exceedingly insignificant compared to the forward current.
Breakdown voltage: The maximum reverse voltage that the diode can withstand without damage. Exceeding this voltage can cause the diode to break down and conduct a large current.

When forward bias diode suddenly reverse biased

In the forward-bias state, there are many electrons from the n -type material progressing through the p -type material and many holes in the n -type material—a requirement for conduction. The electrons in the p-type material and holes progressing through the n-type material establish a large number of minority carriers in each material. If the applied voltage is suddenly reversed to establish a reverse-bias situation, we would ideally like to see the diode change instantaneously from the conduction state to the non-conduction state. However, because of the considerable number of minority carriers in each material, the diode current will simply reverse and stay at this measurable level for the period t_s (storage time) required for the minority carriers to return to their majority-carrier state in the opposite material. The diode will remain in the short-circuit state with a current $I_{reverse}$ determined by the network parameters. Eventually, when this storage phase has passed, the current will be reduced in level to that associated with the non-conduction state. This second period is denoted by t_t (transition interval). The reverse recovery time is the sum of these two intervals: $t_{rrt} = t_s + t_t$



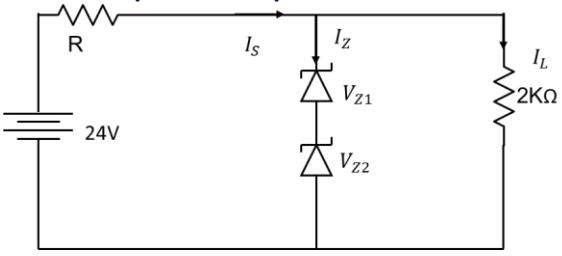
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4. Distinguish between Zener breakdown and avalanche breakdown. The circuit shown uses two Zener diodes, each rated at 10V, 180mA. If the circuit is connected to a 24V unregulated power supply. Determine:

i. The value of series resistance(R)

ii. Maximum power dissipation across each Zener diode.



Solution:

Let us delve into the differences between Zener breakdown and avalanche breakdown in the context of semiconductor diodes:

Zener Breakdown

Mechanism: Zener breakdown occurs due to the high electric field across the PN junction.

Process: When a diode is reverse biased, the kinetic energy of electrons increases, causing them to move at high velocities. These high-velocity electrons collide with other atoms, leading to the creation of free electrons. These newly generated free electrons contribute to a high reverse saturation current.

Voltage Range: Typically observed in Zener diodes with a Zener breakdown voltage of 5 to 8 volts.

Characteristics: Zener breakdown is reversible and does not damage the diode.

Application: Zener diodes are designed to operate in the breakdown region without harm.

Avalanche Breakdown:

Mechanism: Avalanche breakdown occurs due to the collision of free electrons with atoms in the semiconductor material.

Process: When a high reverse voltage is applied across the diode, the electric field across the junction intensifies. This electric field exerts a force on electrons, freeing them from covalent bonds. The liberated electrons gain high velocities and collide with other atoms, creating even more free electrons. This rapid increase in net current characterizes avalanche breakdown.

Voltage Range: Observed in Zener diodes with a Zener breakdown voltage greater than 8 volts.

Characteristics: Avalanche breakdown is irreversible and may permanently damage the diode.

Application: Commonly seen in high-voltage power devices.

Given,

Zener diodes are rated at 10V, 180mA

Supply voltage (V) = 24V

Assume that Zener diode is on with 0 current.

Therefore, $I_L = I_S = \frac{V_L}{R_L} = \frac{20}{2} = 10mA$

So, $R_S = \frac{V_S}{I_S} = \frac{V-V_L}{2} = \frac{4}{10} = 0.4K\Omega$

Now, assume that Zener diode is on with 180mA.

Therefore, $I_S = 180 + I_L = 180 + \frac{V_L}{R_L} = 180 + \frac{20}{2} = 190mA$

So, $R_S = \frac{V_S}{I_S} = \frac{V-V_L}{2} = \frac{4}{190} = 21.05\Omega$

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Maximum power dissipated in each Zener diode

$Power = V_Z \cdot I_Z = 10 \cdot 180 = 1.8W$

5. What is a bipolar junction transistor? Why is it so called? Explain the working principle of NPN transistor with necessary diagram.

Solution:

A Bipolar Junction Transistor (BJT) is a three-terminal semiconductor device made up of three layers of doped semiconductor material. The three layers have alternating p-type and n-type doping, creating two p-n junctions. The three terminals are called:

Base: The thin middle layer.

Emitter: One of the outer layers, heavily doped.

Collector: The other outer layer, moderately doped.

Why is it called "bipolar"

BJTs (Bipolar Junction Transistors) are called bipolar because the conduction of current through them relies on the movement of two types of charge carriers:

Electrons: Negatively charged carriers that primarily flow from the emitter to the collector.

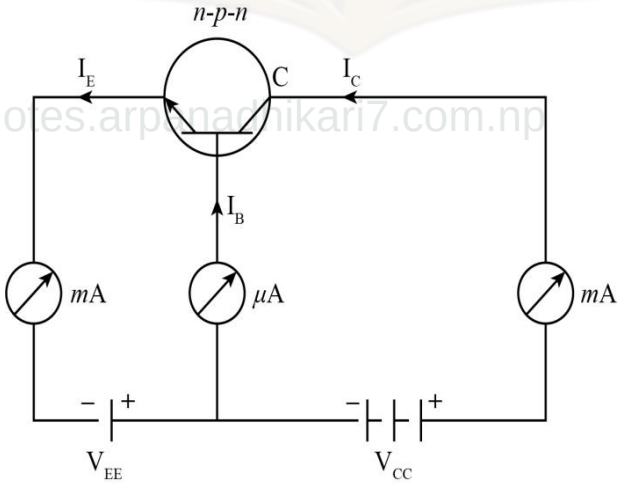
Holes: Positively charged vacancies in the semiconductor lattice that move in the opposite direction of electron flow.

An NPN transistor consists of a thin layer of P-type semiconductor sandwiched between two N-type semiconductor regions. The three sections are named:

Emitter (N): Heavily doped with moderate size.

Base (P): Thin and lightly doped.

Collector (N): Moderately doped with huge size.



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Working Principle

The NPN transistor operates on the principle of controlling the flow of electrons (majority carriers) from the emitter to the collector, using a small current flowing into the base.

Here's a breakdown of the steps in a common emitter configuration:

Forward Bias at the Emitter-Base Junction

The emitter-base junction is forward biased, meaning the positive terminal of the battery (V_{BE}) is connected to the N-type emitter, and the negative terminal is connected to the P-type base. This forward bias lowers the potential barrier at the junction, allowing most electrons from the heavily doped emitter to flow into the thin base region.

Diffusion of Electrons Across the Base

The base is very thin and lightly doped, so only a small percentage of the electrons from the emitter combine with holes in the base region. Most of the electrons diffuse across the base to the collector-base junction.

Reverse Bias at the Collector-Base Junction

The collector-base junction is reverse biased, meaning the positive terminal of the battery (V_{CE}) is connected to the N-type collector and the negative terminal to the P-type base. Despite the reverse bias, the strong electric field created attracts most of the electrons that have diffused to the edge of the depletion region. These electrons are swept into the collector region.

Collector Current

The considerable number of electrons entering the collector gives rise to the collector current (I_C). Since only a few electrons are lost in the base, the collector current is almost equal to the emitter current (I_E).

Key Points

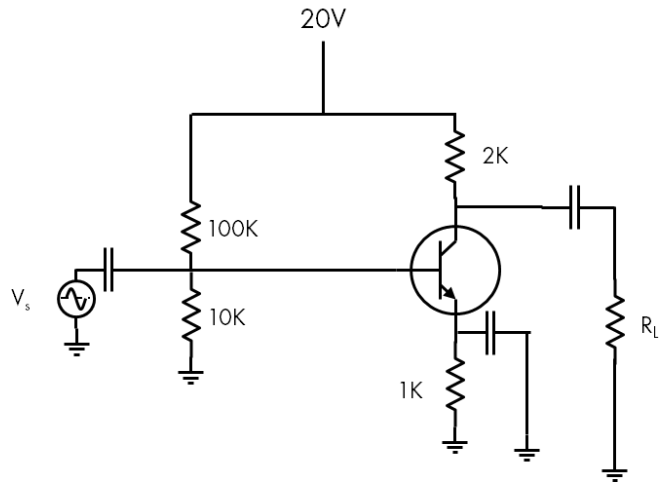
Base current controls collector current: A small current flowing into the base (I_B) can control a much larger current flowing from the emitter to the collector (I_C). This provides the amplification property of the transistor.

The transistor acts like a current-controlled current source.

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6. For the circuit shown below, draw the dc load line and find the operating point. Also find the stability of the circuit. (Assume silicon transistor with β = 100)



Solution:

Given,

$R_1 = 100K\Omega$

$R_2 = 10K\Omega$

$R_C = 2K\Omega$

$R_E = 1K\Omega$

For dc analysis,

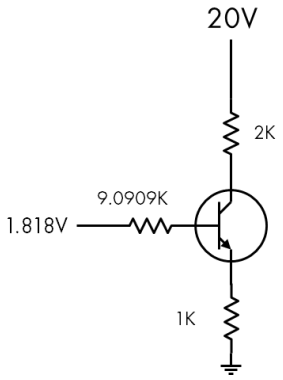
For DC analysis

1. All capacitors are open circuited.
2. All AC sources are grounded.

Now, to find Thevenin equivalent circuit

$$R_{th} = \frac{R_1 * R_2}{R_1 + R_2} = \frac{100 \cdot 10}{100+10} = 9.0909K\Omega$$

$$V_{th} = \frac{R_2 \cdot V_{cc}}{R_1 \cdot R_2} = \frac{10 \cdot 20}{100+10} = 1.818V$$



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Now, using KVL in base to emitter junction,

$V_{th} - R_{th}I_B - V_{BE} - I_E R_E = 0$

$1.818 - 9.0909I_B - 0.7 - \beta I_B = 0$

$I_B = \frac{1.818-0.7}{9.0909+100} = 10.24\mu A$

$I_C = \beta I_B = 100 \cdot 10.24 = 1.024mA$

Now, using KVL in collector to emitter junction,

$V_{cc} - I_C R_C - V_{CE} - I_E R_E = 0$

$20 - 1.024 \cdot 2 - V_{CE} - 1.024 = 0$

$V_{CE} = 20 - 1.024 \cdot 2 - 1.024 = 16.928V$

7. An NPN silicon transistor having a nominal β of 100 is to be used in a CE configuration with $V_{CC} = 15V$. The Q point is to be $I_C = 4mA$ and $V_{CE} = 12V$. Now design the circuit diagram.

Solution:

Given,

$V_{CC} = 15V$

$I_C = 4mA$

$V_{CE} = 12V$

$\beta = 100$

$I_B = \frac{I_C}{100} = 40\mu A$

While designing voltage divider bias, generally, $V_E = \frac{V_{CC}}{10} = \frac{15}{10} = 1.5V$ is set.

Therefore,

$R_E = \frac{V_E}{I_E} = \frac{1.5}{4} = 0.375K\Omega$

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$R_C = \frac{V_{CC}-V_{CE}}{I_C} = \frac{15-12}{4} = 0.75K\Omega$

$V_B = V_{BE} + V_E = 0.7 + 1.5 = 2.1V$

For the circuit to operate efficiently, it is assumed that the current through R1 and R2 should be approximately equal to and much larger than the base current (at least 10:1). This fact and the voltage-divider equation for the base voltage provide the two relationships necessary to determine the base resistors. That is,

$R_2 \leq \frac{1}{10}\beta R_E$

$V_B = \frac{R_2}{R_1+R_2}V_{CC}$

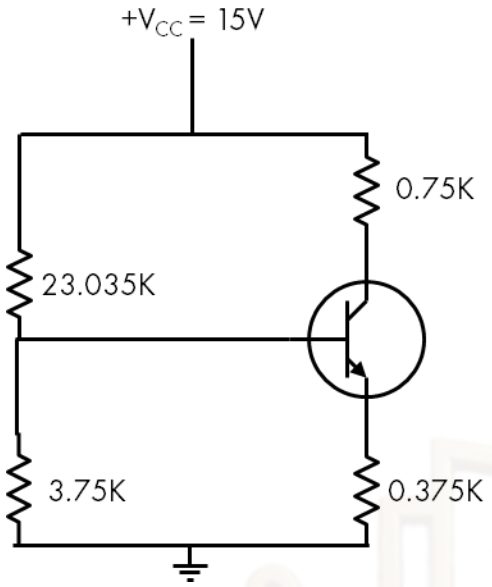
$R_2 = \frac{1}{10}\beta R_E = \frac{1}{10} \cdot 100 \cdot 0.375 = 3.75K\Omega$

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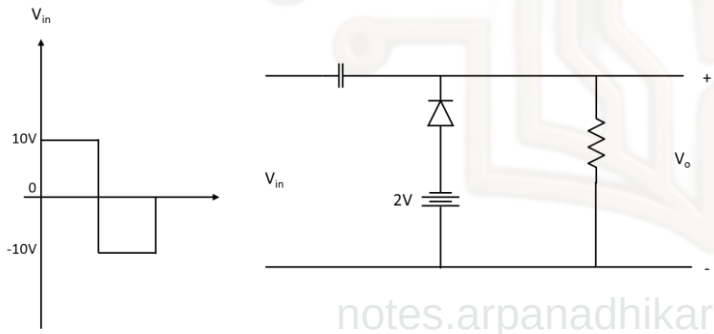
$$2.1 = \frac{3.75}{R_1+3.75}15$$

$$R_1 + 3.75 = 3.75 \cdot \frac{15}{2.1}$$

$$R_1 = 26.785 - 3.75 = 23.035K\Omega$$



8. Draw the output waveform (assume silicon diode).



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Solution:

Input signal is square wave voltage signal with +10V maximum and –10V minimum voltage.

In given circuit,
Diode is on for negative half cycle.
Using KVL in input and battery loop,

$$-V_{in} + V_c - V_D + 2 = 0$$

$$V_c = V_{in} - 2 = 8V$$

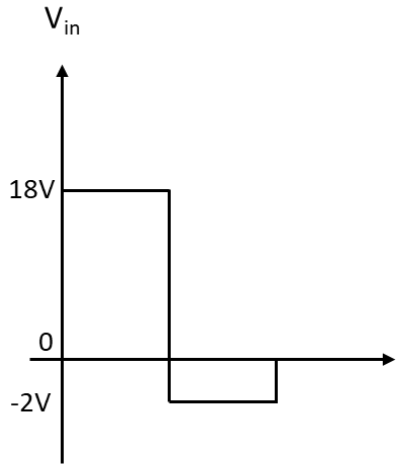
$$V_o = -2V$$

When diode is on

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Now, for positive half cycle,

$V_o - V_c - V_{in} = 0$
$V_o = 8 + 10 = 18V$



9. An L section filter with L = 2H and C = 40μF is used in the output of full wave single phase rectifier that is fed from a 40-0-40V peak transformer. The load current is 0.2A. Calculate the ripple factor. Assume that the rectifier diodes are ideal. Show the circuit diagram.

Solution:

Given,

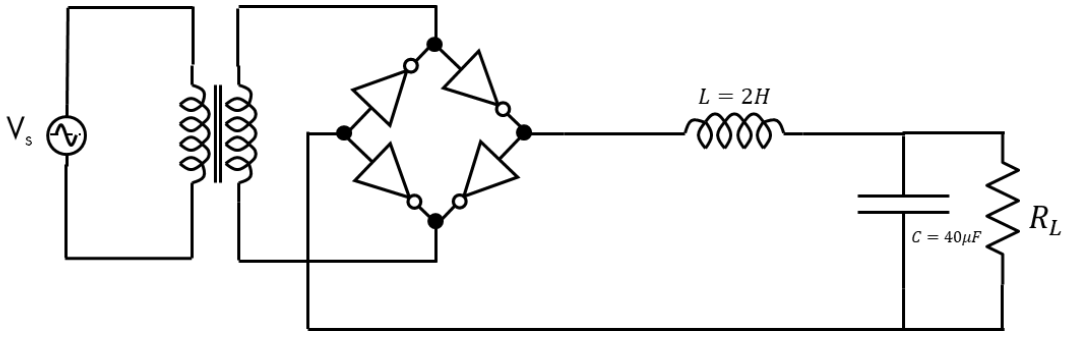
L-section filter with $L = 2H$ and $C = 40\mu F$

Output of transformer is 40-0-40V peak

Load current, $I = 0.2A$

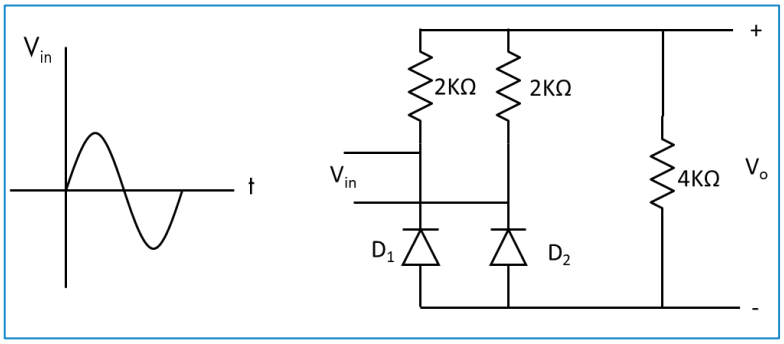
For LC filter, ripple factor,

$$r = \frac{1}{6\sqrt{2}\omega^2 LC} = \frac{1}{6\sqrt{2}(2\pi f)^2 LC} = \frac{1}{6\sqrt{2}(2 \cdot 3.1415 \cdot 50)^2 \cdot 2 \cdot 40 \cdot 10^{-6}} = 0.01493$$



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10. Calculate output voltage and PIV rating of each diode for the circuit shown, with input signal V_{in} = 100Sinωt V.



Solution:

Given,

V_{in} is sinusoidal signal $V_{in} = 100sin\omega tV$

Let's assume that D_2 and D_1 are in on state for positive and negative half-cycles respectively.

For positive half cycle,

$V_{in} = 2i + (2 \parallel 4)i$, Where i be the current following through the circuit

For negative half cycle,

$V_{in} = 2i + (2 \parallel 4)i$, Where i be the current following through the circuit

Since, total resistance in both cases are same i.e ($r_{eq} = 2 + (2 \parallel 4)K\Omega$) current will be same.

Output voltage $V_o = \frac{2\|4}{2+2\|4} V_{in} = \frac{1.333}{2+1.333} \cdot 100sin\omega t = 39.99sin\omega t$

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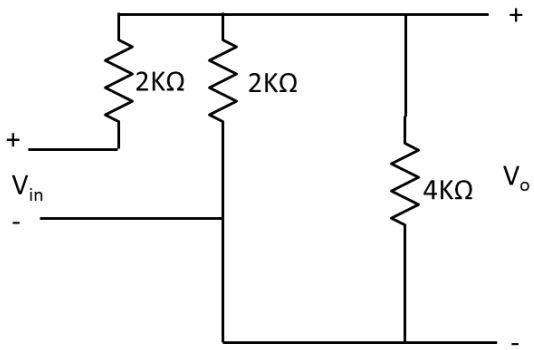


Fig: Positive half cycle

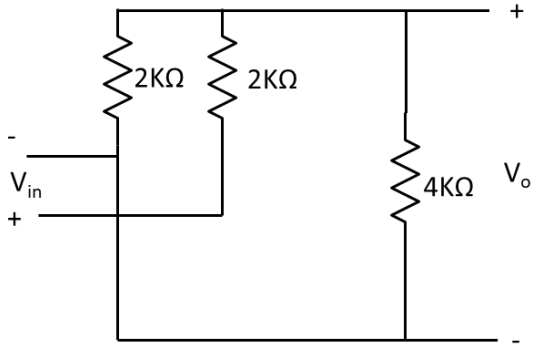
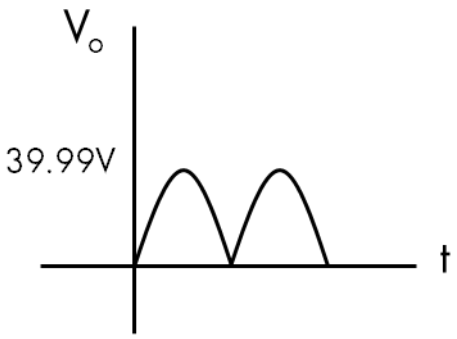


Fig: Negative half cycle

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11. Explain construction and working principle of JFET. Also draw drain and transfer characteristics of a JFET.

Solution:

Construction

N-type Channel: The core of an nJFET is a narrow bar of N-type semiconductor material, usually silicon. This forms the conductive channel between the source and drain terminals.

P-type Gates: Two regions of P-type semiconductor are diffused into the sides of the N-type channel. These regions are called the gates and form P-N junctions with the channel.

Terminals:

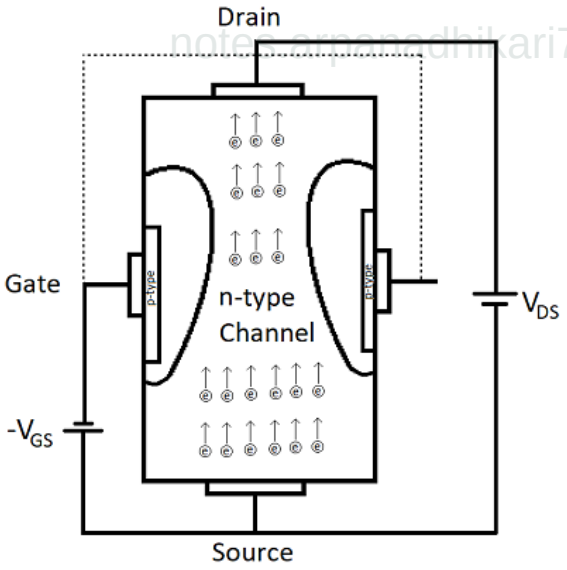
Source: The terminal where charge carriers (electrons in this case) enter the channel.

Drain: The terminal where charge carriers leave the channel.

Gate: The terminal that controls the flow of current through the channel.

Working Principle

$V_{GS} = 0\,V$ **(Zero Gate Voltage):**



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When the gate voltage (V_{GS}) is set to 0 V, the depletion regions around the P-N junctions are still present but have a minimal width.

These depletion regions offer some resistance to the flow of current, but the channel remains wide. This allows a significant amount of current (drain current, I_D) to flow from the source to the drain, limited by the inherent resistance of the channel and the applied drain-source voltage (V_{DS}). In this state, the nJFET operates in the linear region. The drain current increases proportionally with the drain-source voltage.

$V_{GS} < 0\,V$ (Negative Gate Voltage):

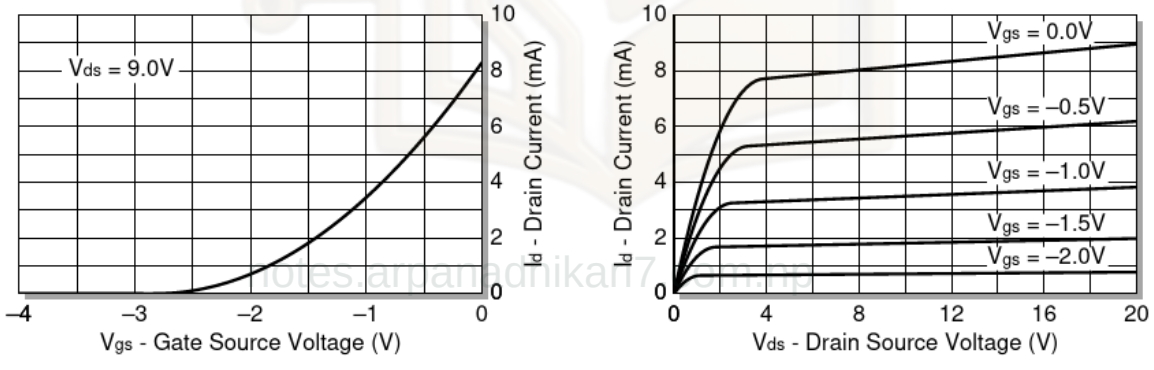
Applying a negative voltage (reverse bias) to the gate with respect to the source widens the depletion regions around the P-N junctions. This widening constriction occurs because the negative voltage attracts the majority charge carriers (electrons) in the N-type channel towards the gate, depleting them from the region closest to the gate.

As the negative gate voltage increases:

The depletion regions become wider, further constricting the channel, and increasing the resistance between source and drain. This results in a decrease in drain current (I_D). Eventually, at a specific negative voltage called the pinch-off voltage (V_p), the depletion regions completely pinch off the channel, leaving no room for current to flow.

Beyond this point, increasing the drain-source voltage (V_{DS}) has minimal effect on the drain current as the channel is already completely pinched off. This region is called the saturation region.

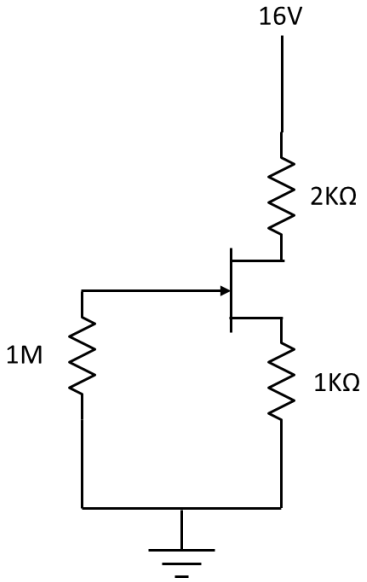
When$V_{GS} = 0\,V$, the nJFET operates in the linear region with a high drain current. As V_{GS}becomes increasingly negative, the drain current progressively decreases due to channel constriction until it reaches zero at the pinch-off voltage, entering the saturation region.



(a) transfer characteristic (b) i-v characteristic

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12. Determine I_{DQ} and V_{DSQ} for the circuit given below. Given I_{DSS} = 8mA, V_p = -8V.



Solution:

Given,

$V_{DD} = 16V$
$R_D = 2K\Omega$
$R_S = 1K\Omega$
$R_G = 1M\Omega$
$V_P = -8V$
$I_{DSS} = 8mA$

Using KVL in gate to source loop,

$$I_G R_G - V_{GS} - I_S R_s = 0$$

Since, $I_G = 0$

$$V_{GS} = -I_S R_s$$

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Since, $I_S = I_D$

$$V_{GS} = -I_D R_s$$

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$I_D = I_{DSS}\left(1 - \frac{-I_D R_S}{V_P}\right)^2$$

$$I_D = I_{DSS}\left(1 + \frac{I_D R_S}{V_P}\right)^2$$

$$I_D = 8\left(1 - \frac{I_D}{8}\right)^2$$

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$8I_D = (8 - I_D)^2$
$I_D^2 - 24I_D + 64 = 0$

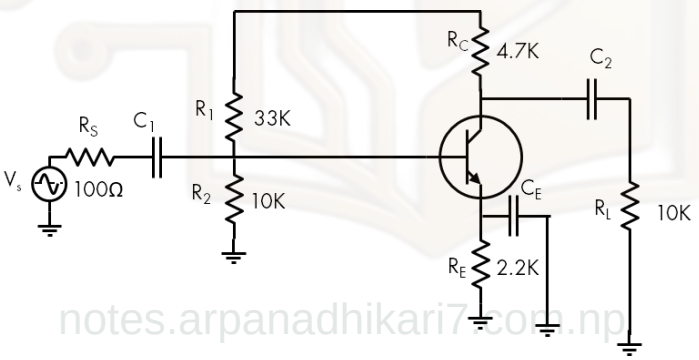
Solving above equation,
$I_D = 3.0557\ mA, 20.9443mA$
Since 20.9443 mA is greater than $I_{DSS} = 8mA$, we choose 3.0557 mA.

Now, Using KVL in drain to source loop,
$V_{DD} - I_D R_D - V_{DS} - I_S R_s = 0$
$16 - 3.0557 \times 2 - V_{DS} - 3.0557 = 0$
$V_{DS} = 6.829V$

Therefore, $V_{DSQ} = 6.829V$ and $I_{DQ} = 3.0557\ mA$

13. The transistor shown in figure below has the following h-parameter value $h_{ie} = 1K\Omega$, $h_{re} = 2.5 \times 10^{-4}$, $h_{fe} = 50$, $h_{oe} = 25\mu s$. Find:

I. Input impedance
II. Voltage gain
III. Current gain
IV. Output impedance



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Solution:

Given,

$h_{ie} = 1K\Omega$

$h_{re} = 2.5 \times 10^{-4}$

$h_{fe} = 50$

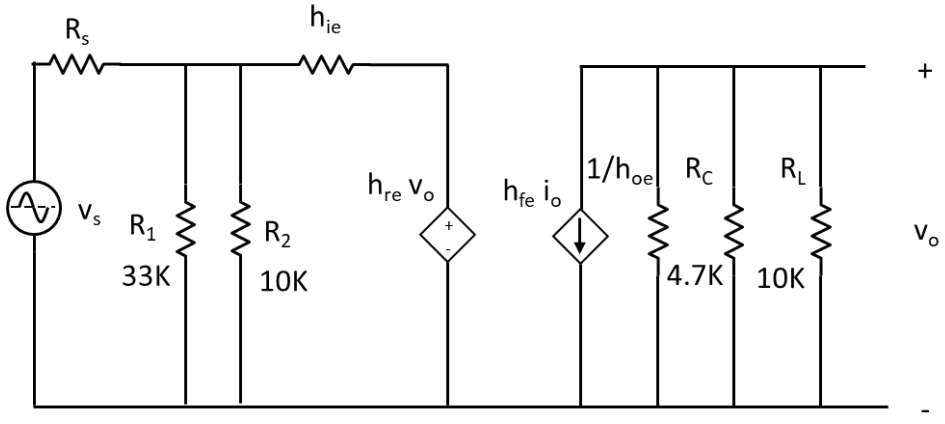
$h_{oe} = 25\mu s$

We know, for two port networks with h-parameter

$$v_i = h_{ie} i_i + h_{re} v_o$$
$$i_o = h_{fe} i_i + h_{oe} v_o$$

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$v_o = -i_o R_L$ where R_L = 10//4.7 = 3.197KΩ



Current gain (A_i) = $\frac{h_{fe}}{1+h_{oe}R_L} = \frac{50}{1+0.025 \cdot 3.197} = 46.299$

Input Impedance, $Z_i = \frac{v_i}{i_i} = h_{ie} - \frac{h_{fe}h_{re}R_L}{1+h_o R_L} = 1 - \frac{50 \cdot 2.5 \cdot 10^{-4} \cdot 3.197}{1+0.025 \cdot 3.197} = 0.963K\Omega$

$$Z_i' = 0.963 \,||33||10 \; = 0.8557K\Omega$$

Voltage gain (A_v) = $\frac{v_o}{v_i} = \frac{-h_{fe}R_L}{h_{ie}+(h_i h_o - h_f h_r)R_L} = \frac{50 \cdot 3.197}{1+(1 \cdot 0.025 - 50 \cdot 2.5 \cdot 10^{-4})3.197} = 153.707$

Output Impedance, $Z_o' = \frac{v_o}{i_o} = \frac{1}{h_{oe} - \left[\frac{h_{fe}h_{re}}{h_{ie}+R_s}\right]} = \frac{1}{0.025 - \left(\frac{50 \cdot 2.5 \cdot 10^{-4}}{1+0.1}\right)} = 73.333K\Omega$

$$Z_o = R_C||Z_o' = 4.7||73.333 = 4.416K\Omega$$

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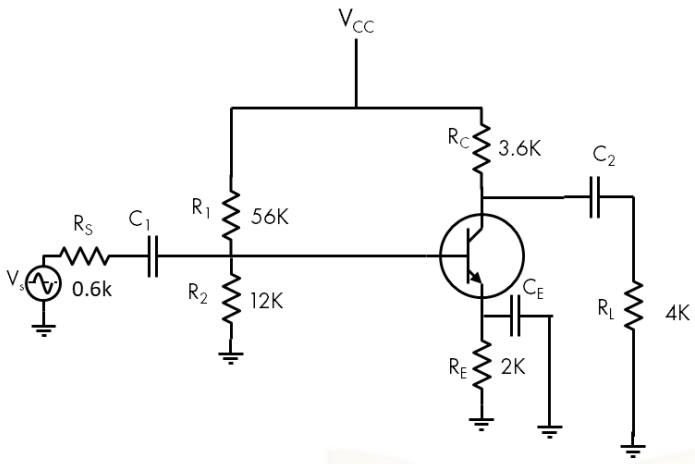
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14. For the given CE amplifier, find the following using r_e model

i. Input Impedance

ii. Output impedance

iii. Voltage gain (Assume β = 90, Vcc = +19V)



Solution:

$R_1 = 56K\Omega$

$R_2 = 12K\Omega$

$R_C = 3.6K\Omega$

$R_E = 2K\Omega$

$R_S = 0.6K\Omega$

$R_L = 4K\Omega$

To find r_e, DC analysis should be done.

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For DC analysis

1. All capacitors are open circuited.
1. All AC sources are grounded.

Now, to find Thevenin equivalent circuit

$$R_{th} = \frac{R_1 + R_2}{R_1 \cdot R_2} = \frac{56 \cdot 12}{56 + 12} = 9.882K\Omega$$

$$V_{th} = \frac{R_2 \cdot V_{cc}}{R_1 \cdot R_2} = \frac{12 \cdot 19}{56 + 12} = 3.3529V$$

Now, using KVL in base to emitter junction,

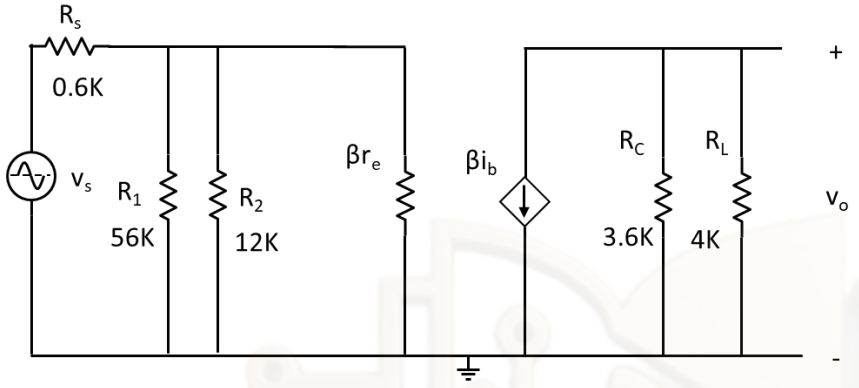
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$V_{th} - R_{th}I_B - V_{BE} - I_E R_E = 0$

$$3.3529 - \frac{9.882 I_E}{\beta} - 0.7 - I_E \cdot 2 = 0$$

$$I_E = \frac{3.3529 - 0.7}{2 + \frac{9.882}{90}} = 1.257mA$$

Therefore, $r_e = \frac{26mV}{1.257mA} = 20.684\Omega$



Input impedance, $Z_i = \beta r_e || 56 || 12 = \left(\frac{1}{56} + \frac{1}{12} + \frac{1}{0.09 * 20.684}\right)^{-1} = 1.566K\Omega$

Output impedance, $Z_o = R_c = 3.6K\Omega$

Voltage gain $A_{vsl} = \frac{V_l}{V_s} = \frac{V_l}{V_{in}} \cdot \frac{V_{in}}{V_s} = -\frac{i_c(R_C || R_L)}{i_b \beta r_e} \cdot \frac{Z_i}{Z_i + R_s} = \frac{3.6||4}{0.020684} \cdot \frac{1.566}{1.566+0.6}$

$$A_{vsl} = \frac{1.894}{0.020684} \cdot \frac{1.566}{2.166} = 66.203$$

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15. Darlington-pair amplifier is called super beta, why? Derive the necessary expression of its efficiency.

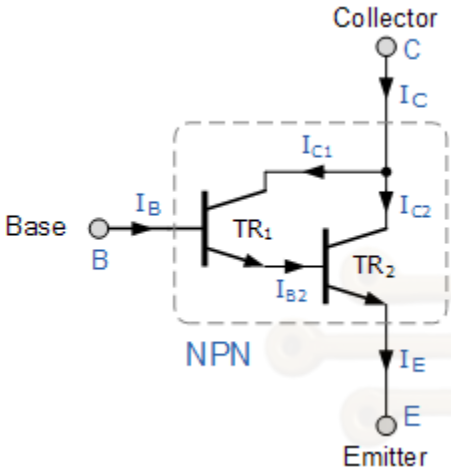
Solution:

The Darlington pair, is a configuration of two bipolar junction transistors (BJTs) connected in a specific way to achieve several advantages, including:

High current gain (beta): This is the primary reason it's called "super beta." The overall current gain of the Darlington pair is significantly higher than that of an individual transistor.

High input impedance: This makes the circuit more sensitive to input current.

In a Darlington pair, the collector current of the first transistor (Q1) acts as the base current for the second transistor (Q2). Since the current gain of each transistor is greater than 1, the overall current gain of the pair is the product of the individual gains.



Assuming a beta β_1 for the first transistor and β_2 for the second, the base current for the second transistor is,

$$I_{B2} = I_{E1} = (\beta_1 + 1)I_{B1}$$

And the emitter current for the second transistor is

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$$I_{E2} = (\beta_2 + 1)I_{B2} = (\beta_1 + 1)(\beta_2 + 1)I_{B1}$$

Assuming $\beta \gg 1$ for each transistor, we find the net beta for the configuration is,

$$\beta_D = \beta_1\beta_2$$

This product can be significantly larger than the β of a single transistor, making the Darlington pair a "super beta" amplifier.

Efficiency of a Darlington Pair:

The efficiency of an amplifier refers to the ratio of the output power delivered to the load to the total power supplied to the amplifier. Calculating the exact efficiency of a Darlington pair amplifier circuit requires specific details about the circuit configuration, transistor characteristics, and operating conditions. However, we can derive a general expression for the efficiency:

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η ≈ (Pout / Pin) * 100%

where:

η is the efficiency (as a percentage)

Pout is the output power delivered to the load

Pin is the total power supplied to the amplifier

Pout can be calculated using the following equation for a BJT amplifier in the saturation region:

$$P_{out} = V_{CC} \cdot Current\ drawn$$

where:

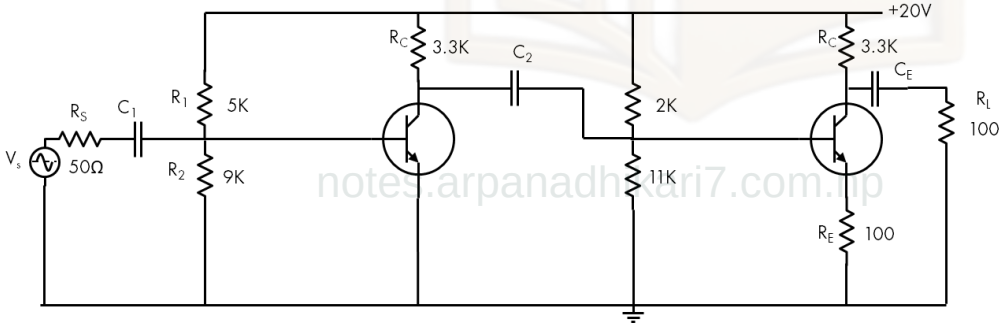
Vcc is the driving voltage

Current drawn depends on the operating point.

Pin can be estimated by considering the power supply voltage (Vcc) and the total current drawn by the amplifier (Itotal). However, calculating the exact Itotal requires further analysis of the specific circuit configuration and component values.

Therefore, providing a specific numerical value for the efficiency is not possible without further information about the circuit. However, the provided expressions offer a general understanding of how the efficiency of a Darlington pair amplifier can be determined.

16. Find the overall voltage gain of cascaded transistor amplifier as shown in the figure below. $r_{e1} = r_{e2} = 25\Omega$ **and** $\boldsymbol{\beta 1 = \beta 2 = 100}$ $\boldsymbol{\beta 1 = \beta 2 = 100.}$



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Solution:

Given,

$r_{e1} = r_{e2} = 25\Omega$

$\beta_1 = \beta_2 = 100$

$R_1 = 5K\Omega$

$R_2 = 9K\Omega$

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$R_{C1} = R_{C2} = 3.3K\Omega$

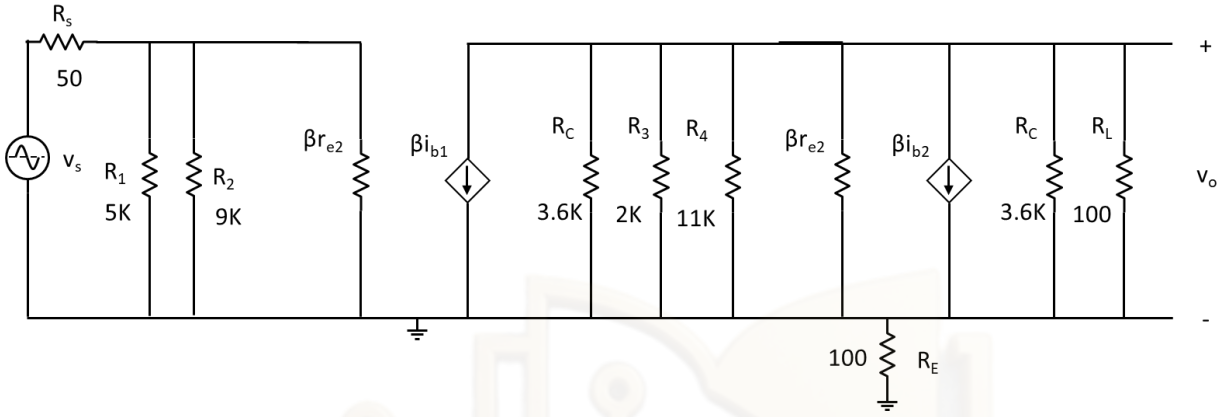
$R_3 = 2K\Omega$

$R_4 = 11K\Omega$

$R_{E2} = 100\Omega$

$R_L = 100\Omega$

$R_S = 50\Omega$



Now, input impedance of first stage,

$$Z_{i1} = 5||9||\beta r_{e1} = \left(\frac{1}{5} + \frac{1}{9} + \frac{1}{2.5}\right)^{-1} = 1.406K\Omega$$

Output impedance of first stage,

$$Z_{o1} = R_C||2||11||\beta(r_{e2} + R_{E2}) = \left(\frac{1}{3.6} + \frac{1}{2} + \frac{1}{11} + \frac{1}{0.1(100+25)}\right)^{-1} = 1.054K\Omega$$

Input impedance of second stage,

$$Z_{i2} = 2||11||\beta(r_{e2} + R_{E2}) = \left(\frac{1}{2} + \frac{1}{11} + \frac{1}{0.1(25+100)}\right)^{-1} = 1.49K\Omega$$

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Output impedance of second stage,

$$Z_{o2} = R_C = 3.6$$

Overall voltage gain,

$$A_{vsl} = \frac{V_l}{V_s} = \frac{V_l}{V_{in2}} \cdot \frac{V_{o1}}{V_{in1}} \cdot \frac{V_{in1}}{V_s}$$

$$A_{vsl} = -\frac{i_{c2}(R_C\,||R_L)}{i_{b2}\beta r_e + i_{e2}R_E} \cdot \frac{-i_{c1}Z_{o1}}{i_{b1}\beta r_e} \frac{Z_{i1}}{Z_{i1} + R_s} = \frac{3.6||0.1}{0.125} \cdot \frac{1.054}{0.025} \cdot \frac{1.406}{1.406 + 0.05}$$

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$$A_{vsl} = \frac{0.0973}{0.125} \cdot \frac{1.054}{0.025} \cdot \frac{1.406}{1.406 + 0.05} = 31.69$$

17. Explain the advantages and disadvantages of negative feedback. Negative feedback of 0.2% is applied to an amplifier of gain 60 dB. Calculate the percentage change in the overall gain of the feedback amplifier if the internal amplifier is subjected to gain reduction of 15%.

Solution:

Advantages of Negative Feedback Amplifiers

Increased Stability: Negative feedback reduces the overall gain of the amplifier, making it less sensitive to variations in component values, temperature changes, and aging effects. This leads to a more stable and predictable gain performance.

Reduced Distortion: Non-linarites in the amplifier circuit can cause distortion in the output signal. Negative feedback helps to cancel out unwanted harmonics and higher-order products, resulting in a cleaner and less distorted output signal.

Improved Bandwidth: In specific configurations, negative feedback can help to widen the bandwidth of the amplifier, allowing it to amplify a wider range of frequencies.

Lower Output Impedance: Negative feedback can decrease the output impedance of the amplifier, making it behave more like an ideal voltage source. This can be beneficial for driving loads that require low impedance.

Disadvantages of Negative Feedback Amplifiers

Reduced Gain: As mentioned earlier, negative feedback decreases the overall gain of the amplifier. This may require additional stages of amplification to achieve the desired output level.

Increased Design Complexity: Implementing negative feedback requires additional circuitry, making the amplifier design slightly more complex compared to non-feedback designs.

Potential for Oscillation: Improper design and component selection with negative feedback can lead to oscillations in the amplifier, resulting in unwanted output signals.

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Given, Open loop gain in decibel $A_{dB} = 60dB$

$$20 \log_{10} A = 60$$

$$\log_{10} A = 3$$

$$A = 10^3 = 1000$$

$\frac{dA}{A} = 15\%$ and $\beta = 0.2\% = 0.002$

Now, the percentage change in the gain after negative feedback is given by,

$$\frac{dA_f}{A_f} = \frac{dA}{A} \cdot \frac{1}{1 + A\beta}$$

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$$\frac{dA_f}{A_f} = 15\% \frac{1}{1 + 1000 \cdot 0.002} = 5\%$$

Therefore, the % change in overall gain of feedback amplifier is 5%.

18. Calculate voltage gain, input, and output impedance with feedback for voltage series feedback having $A = 300, R_i = 1.5K\Omega, R_0 = 50K\Omega$ and $\beta = 1/15$.

Solution:

Given,

$$A = 300$$

$$R_i = 1.5K\Omega$$

$$R_o = 50K\Omega$$

$$\beta = \frac{1}{15}$$

We know,

$$Gain\ after\ feedback\ (A_f) = \frac{1}{1 + A\beta} = \frac{1}{1 + 300 \cdot \frac{1}{15}} = 0.0476$$

$$Input\ impedence\ after\ feedback\ (R_{if}) = R_i(1 + A\beta) = 1.5\left(1 + 300 \cdot \frac{1}{15}\right) = 31.5K\Omega$$

$$Output\ impedence\ after\ feedback\ (R_{of}) = \frac{R_o}{(1 + A\beta)} = \frac{50}{1 + 300 \cdot \frac{1}{15}} = 2.38K\Omega$$

19. Write the salient feature of operational amplifier. Design a summing operational amplifier circuit that will produce an output voltage $V_{out} = -4V_1 - V_2 + 0.1V_3$

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Solution:

An operational amplifier, commonly known as an op-amp, is a fundamental building block in analog electronic circuits. Let us explore its key features:

- **High Gain**: Op-amps exhibit high voltage amplification. They can significantly boost the input signal to produce a larger output signal. This high gain makes them suitable for various applications.
- **Differential Voltage Amplifier**: An op-amp is a two-input, single-output differential voltage amplifier. It takes two input signals (usually labeled as V1 and V2) and produces an amplified output based on their difference.
- **Input Impedance**: Op-amps have extremely high input impedance. This means they draw minimal current from the input sources, making them ideal for interfacing with sensors and other low-current devices.

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- **Output Impedance**: The output impedance of an op-amp is low, allowing it to drive loads (such as resistors or capacitors) effectively without significant voltage drop.
- **External Feedback Components**: Op-amps are designed to work with external feedback components, such as resistors and capacitors. These components determine the amplifier's behavior and functionality.
- **Inverting and Non-Inverting Inputs**: The inverting input (marked with a negative sign, "-") processes the input signal with inverted polarity. The non-inverting input (marked with a positive sign, "+") processes the input signal without inversion.

Given,

$$V_{out} = -4V_1 - V_2 + 0.1V_3$$

$$V_{out} = -\left(4V_1 + V_2 + 0.1(-V_3)\right)$$

The standard form of summing op-amp circuit is,

$$V_{out} = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3\right)$$

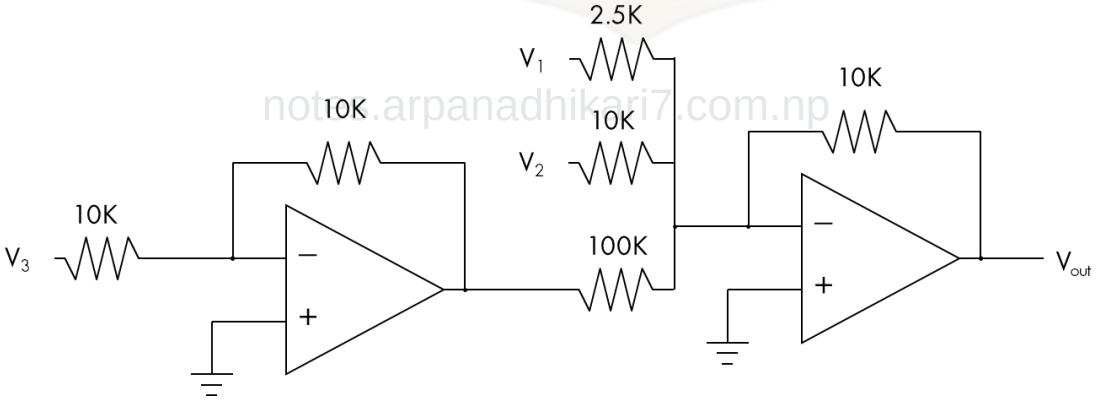
Let's take $R_f = 10K\Omega$

Now,

$$R_1 = \frac{R_f}{4} = \frac{10}{4} = 2.5K\Omega$$

$$R_2 = \frac{R_f}{1} = \frac{10}{1} = 10K\Omega$$

$$R_3 = \frac{R_f}{0.1} = \frac{10}{0.1} = 100K\Omega$$



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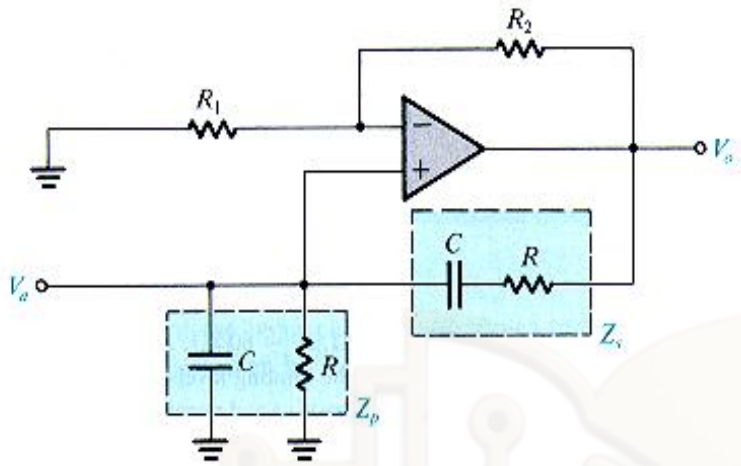
20. For the op-amp based Wein bridge oscillator if the component values are:
$R = 5.1k\Omega, C = 1nF, R_2 = 12K\Omega$ and $R_1 = 5.1K\Omega$

i. Determine whether the circuit will oscillate or not.

ii. Obtain the output frequency.

Solution:

Given,



$$R = 5.1K\Omega,\ C = 1nF,\ R_2 = 12K\Omega\ and\ R_1 = 5.1K\Omega$$

As we know, ideal condition for Wein bridge oscillator to oscillate

$$\frac{R_2}{R_1} = 2$$

From given parameters, $\frac{R_2}{R_1} = \frac{12}{5.1} = 2.353$ is suitable for practical purpose.

Oscillation frequency $f = \frac{1}{2\pi RC} = \frac{1}{2 \cdot 3.14 \cdot 5.1 \cdot 10^3 \cdot 1 \cdot 10^{-9}} = 31.22KHz$

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